

Physis-Lang: Self-Evolving Language as a Physical Representation for Video World Model

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Video world models are expected to predict how the physical world evolves, yet they often produce visually plausible videos that violate basic physical principles. Existing approaches commonly assume that natural language is insufficient to represent the physical knowledge required for reliable generation, and therefore introduce additional visual, latent, numerical, or planning-based signals. We revisit this assumption and introduce *Physis-Lang*, a self-evolving framework that treats physical language as a shared and optimizable representation across data curation, model training, and video generation. Physis-Lang represents physical processes through language that describes their relevant entities, causes, interactions, governing principles, temporal evolution, and effects. To improve this representation, we construct *PhysCapBench*, which decomposes physical processes into atomic assertions and evaluates captions using recall and precision. An agentic loop iteratively analyzes assertion-level errors and refines the instruction used to produce physical captions. Physis-Lang further converts model deficiencies into textual descriptions and uses language-guided retrieval to identify visually diverse videos that cover missing physical processes. Experiments on four widely used physical video benchmarks with Wan and Cosmos backbones demonstrate consistent improvements in physical plausibility. Notably, starting from open-source Cosmos3-Nano backbones, our Physis-Lang-enhanced models surpass the leading proprietary Veo 3.1 model.

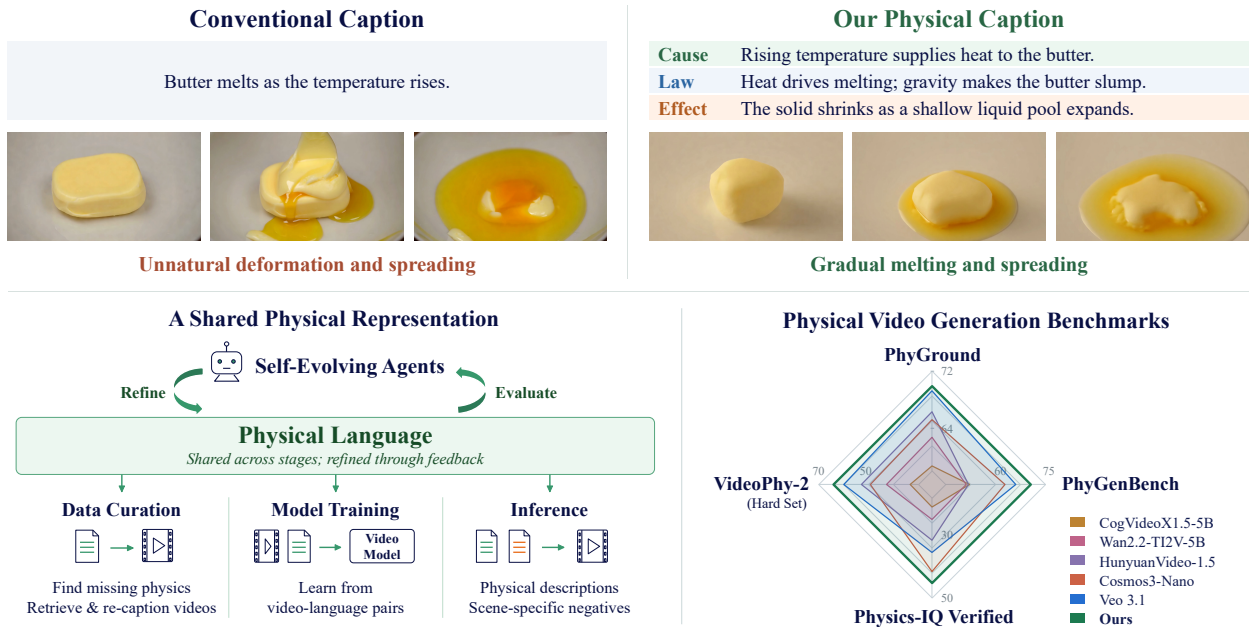


Figure 1 | **Overview of Physis-Lang.** Existing approaches treat language mainly as a conditioning interface and introduce physical knowledge through separate visual, latent, numerical, or planning-based signals. Physis-Lang instead treats physical language as a shared and optimizable representation: an agentic loop evaluates and refines physical descriptions, language-guided retrieval expands the training data toward missing physical processes, and the evolved language supervises training and guides inference toward physically plausible video generation.

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1. Introduction

Video world models are expected not only to synthesize visually appealing videos, but also to predict how the physical world evolves. This capability is fundamental to embodied intelligence, robotics, autonomous systems, and interactive simulation, where a model must anticipate the consequences of actions rather than merely render plausible-looking frames. However, conventional video-text training primarily optimizes visual fidelity and semantic alignment, without explicitly representing the physical knowledge underlying observed dynamics; as a result, current video generation models still frequently violate basic physical principles: objects deform or disappear unexpectedly, collisions produce implausible outcomes, fluids move unnaturally, and causal events occur in the wrong order. Recent benchmarks have further shown that perceptual realism does not necessarily imply physical understanding [1, 2, 3, 4]. Improving the *physical plausibility* of video world models therefore remains a central challenge on the path toward general-purpose world simulation.

Existing efforts improve the physical plausibility of video generation by complementing the standard language interface with additional sources of physical information. They enrich video-text data with physics-focused videos or physical annotations [5, 6], introduce auxiliary signals about geometry, motion, or physical correctness during training [7, 8], and provide structured plans, retrieved examples, or external judgments during generation [9, 10]. Despite their different implementations, these approaches reflect a common design choice: natural language is primarily used to specify semantic content, while more detailed physical information is conveyed through complementary visual, geometric, latent, numerical, or human-designed signals. These complementary signals can provide useful supervision for aspects of physical processes that language alone may not fully capture. Yet an important possibility remains relatively unexplored: *can language itself, when made more explicit and systematically optimized, serve as a unified representation of physical knowledge across the video-generation pipeline?*

Motivated by this question, we hypothesize that the limited role of language in current video models may stem from language not yet being fully exploited as a representation of physical processes, rather than from an intrinsic limitation of language itself. Based on this view, we introduce *Physis-Lang*, a self-evolving agentic framework that constructs, evaluates, and refines physical language as a shared representation, as illustrated in Fig. 1. At the caption level, Physis-Lang goes beyond describing visible actions and makes the relevant physical entities, causal interactions, governing principles, and resulting effects explicit. The self-evolving representation is then reused throughout the pipeline: it guides data curation by identifying missing physical domains and selecting relevant videos, provides supervision for training on video-language pairs, and guides inference through scene-specific physical descriptions. In this way, the three components shown at the bottom of Fig. 1 are connected by the shared physical language at its center, enabling the system to organize, evaluate, and transfer physical knowledge across data curation, training, and inference.

To build this shared representation, we first instantiate it at the caption level through an agentic caption-evolution loop grounded by our *physics-aware critic* and *PhysCapBench*. Given a fixed captioning model and an initial instruction, the model generates candidate physical descriptions for videos in our development set. For each video, we provide human-curated reference assertions organized around *cause*, *physical law*, and *effect*. Using these references, our critic evaluates each candidate description along two dimensions: its coverage of the relevant physical knowledge and the faithfulness of its individual claims to the visual evidence. An evolution agent uses this diagnostic feedback to identify omissions, unsupported claims, and vague causal relations, and then revises the captioning instruction without updating the captioner’s weights. Each instruction is then evaluated on our PhysCapBench within the same loop, which continues until benchmark performance saturates. The best-performing instruction is then used to re-caption the training corpus. Thus, our physics-aware critic provides the loop feedback and PhysCapBench provides stopping criterion for language evolution, while Physis-Lang turns that feedback into improved physical descriptions for video-model training and inference.

The self-evolving physical language also enables Physis-Lang to expand the training distribution toward physical domains that the target video model handles poorly. It first summarizes the model’s failures into a category-level deficiency profile. These deficient domains are then matched against physics tags associated with videos in a large gallery. Unlike retrieval based solely on visual similarity, this language-guided matching selects videos according to their physical content rather than their appearance, allowing the data engine to target missing physical domains and collect visually diverse examples relevant to the same deficiency. The selected videos are re-captioned with the self-evolving guidelines and added to the training corpus, directing additional supervision toward the model’s weaknesses. Together with positive physical reasoning

and scene-specific negative descriptions at inference time, this language-guided data engine connects targeted data curation, model training, and generation through a shared physical-language representation.

Our main contributions are summarized as follows:

- We introduce a new perspective that treats **language as a physical world representation**. We show that language, when explicitly structured and self-evolving, can serve as a representation for carrying, refining, and applying physical knowledge in video world models.
- We develop a **self-evolving agent system** that optimizes physical-language guidelines using a physics-aware critic. We introduce **PhysCapBench** to evaluate physical coverage and claim faithfulness and to guide iterative refinement.
- We build a **language-guided data engine** that uses physics-domain tags and text-level matching to select training videos relevant to model deficiencies, then enriches them with self-evolving physical captions.
- Extensive experiments on four physical video benchmarks demonstrate significant and consistent improvements on both Wan and Cosmos backbones. Our resulting models also outperform Veo 3.1 under the same evaluation protocols, validating language as a practical representation for improving video world models.

2. Related Work

Physics-Aware Video Generation. Recent efforts to enhance the physical plausibility of video generative models can be categorized by the source of the physical priors they exploit. The largest line of work draws physical priors from pretrained vision-language or foundation models, utilizing them to distill implicit physical dynamics into auxiliary branches [11, 12, 7, 13, 14, 15, 16], retrieve physically grounded reference motions [17, 18, 19], verify or critique generated contents [9, 20, 21, 22, 23, 24, 25], or plan over intermediate representations before rendering [26, 27, 28, 29, 30]. A second line treats physical plausibility as an objective to be optimized via reinforcement learning or preference optimization, aligning the generator with reward or preference signals derived from physics-aware judges or contrastive trajectories [31, 32, 33, 34, 35, 36, 37, 38, 39]. A third line of works injects explicit graphics- or simulator-derived structures like physical equations, trajectories, or executable simulation code into the generation pipeline [8, 40, 41, 42, 43, 10, 44, 45, 46, 47]. Some works also curate large-scale physics-annotated datasets [6, 5, 48].

Despite these advances, most existing approaches rely on auxiliary mechanisms, such as reward models, intermediate features, or retrieved samples, to incorporate physical knowledge. Simulation-based approaches, on the other hand, are constrained by high computational costs and limited generalizability. In contrast, we posit that language provides a compact, explicit, and transferable representation of physical knowledge. Building on this view, we introduce an agentic self-evolution prompting scheme that iteratively refines the language representation to better capture physical priors. Different from the views of many recent works, we prove that *language alone* is very powerful to lead to physically plausible video generation if we utilize it properly.

Agentic Self-Evolution. Self-evolution has emerged as a distinct paradigm for improving agents, which has been applied to a range of tasks, including web navigation [49, 50, 51, 52], mathematical and code reasoning [53, 54, 55, 56, 57, 58, 59], general instruction-following and question answering [60, 61], as well as interactive tool use [62, 63, 64]. Across these tasks, an agent’s prompts, policy, memory, or tools are iteratively refined. Representative methods involve revising subsequent attempts through verbal self-critique of past failures [65, 66, 67], using the model itself as a judge to construct preference data for iterative alignment training [68, 69], or training agents via multi-turn reinforcement learning within an interactive environment [70, 71, 72]. In contrast, we are the first to apply agentic self-evolution to physics reasoning that ultimately enhances the physical realism of video generation tasks. We establish a physics caption benchmark to evaluate the agents, which allows the upsampled prompts to converge toward a format that is optimally suited for generating physically realistic videos.

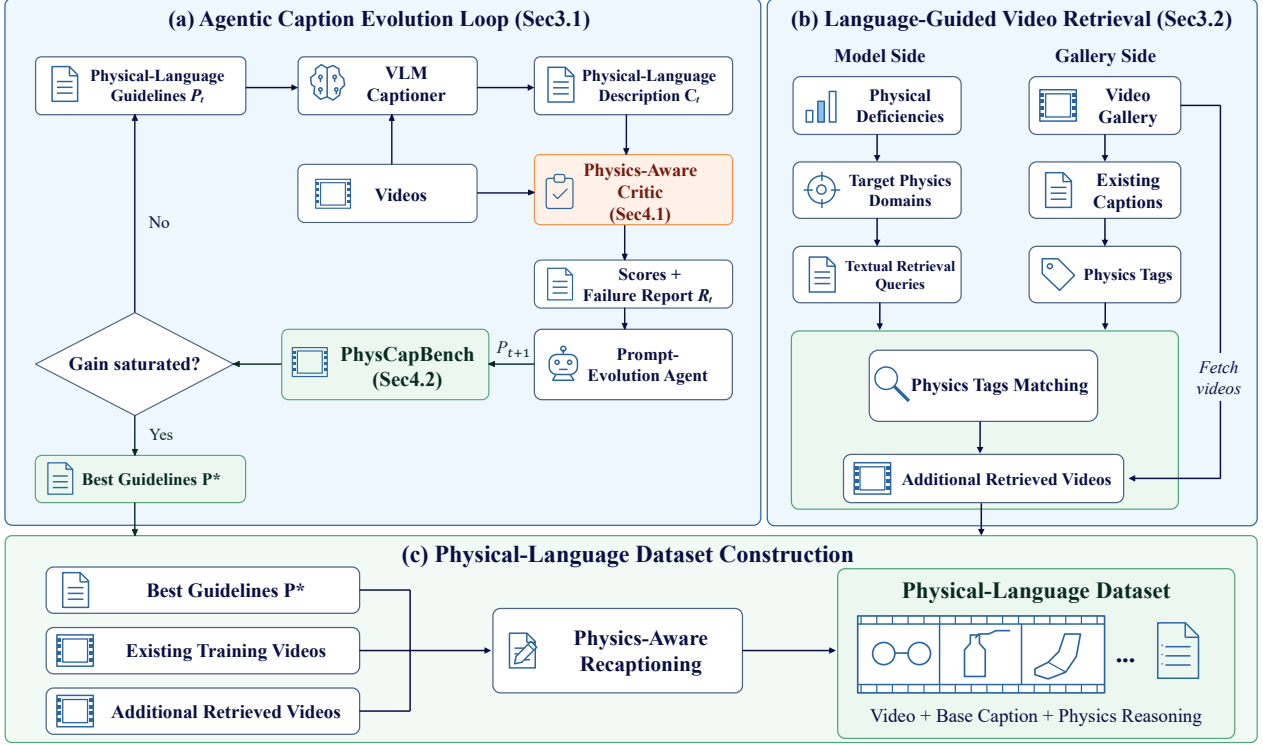


Figure 2 | **Overview of our method.** Physical processes are represented through natural language, including a base caption, explicit physics reasoning, and scene-adaptive negative physics descriptions. A PhysCapBench-driven agentic loop evaluates and refines this physical language, which is then used to expand the training data, supervise video-model training, and guide inference toward physically plausible generation.

3. Method

3.1. Self-Evolving Agents to Improve Physics Caption

Fig. 2(a) demonstrates the overview of our agentic caption evolution loop. Conventional captions often describe scene semantics without explaining physical causes and consequences. We make these mechanisms explicit in language so that video-text training can provide richer supervision for physical dynamics. To this end, we augment base caption with a natural-language *physics_reasoning* field describing physical details, dynamics, and causal relations. The same representation supports training supervision and image/text-conditioned prompt expansion at inference. Because captioning instructions can omit key details or induce unsupported reasoning, we use a self-evolving agent to iteratively refine the instruction while keeping the captioning model fixed.

To enable systematic prompt evolution, we first develop a **physics-aware critic** that evaluates the correctness and completeness of physical descriptions. The detailed protocol is introduced in Sec. 4.1. Based on this critic, we further introduce **PhysCapBench**, a held-out benchmark of 246 physical videos for evaluating physics-aware video captioning. It serves both as a standalone benchmark and as the validation set for prompt evolution.

We then construct an agentic self-evolution loop over a fixed 20-video development set. At iteration t , a fixed captioner uses prompt P_t to generate captions C_t . The critic produces a score and diagnostic report R_t , which summarizes errors and guides an evolution agent to revise the prompt: $P_t \rightarrow C_t \rightarrow R_t \rightarrow P_{t+1}$. After each iteration, the updated prompt is evaluated on PhysCapBench. We continue the loop while benchmark performance improves and stop once the gain saturates, selecting the best-performing prompt as the final physics-captioning instruction. Detailed implementation is provided in App. B.2.

During training, we apply the final captioner to re-caption the collected videos with physics-rich descriptions. During inference, we construct a corresponding upsampler that expands short text prompt and an input

image (optional) into a detailed physics-aware prompt, providing the video generator with more explicit physical guidance and improving the physical realism of the generated video. We further derive a scene-specific ***physics_negative_prompt*** describing likely physically implausible evolutions, which serves as negative conditioning to suppress physical inconsistencies and improve generation realism at inference time.

3.2. Language-Guided Video Retrieval

Beyond improving the linguistic representation of physical processes, we further leverage language as a key modality for targeted data expansion, using it to identify and collect training examples that address the physical deficiencies of pretrained video generative models. Fig. 2(b) illustrates the framework of this data collecting pipeline.

We first build a GPT-5.5-based diagnosis agent to identify physical failures in generated videos and assign each failure to physics categories, such as rigid-body motion, collision and fluid dynamics. Aggregating these results yields a category-level deficiency profile that captures the model’s dominant physical weaknesses. We then utilize this profile to retrieve targeted training data from a large video gallery, where each candidate video is tagged with physics-domain labels derived from its caption, enabling retrieval that matches the deficient categories. The retrieved videos are added to the training set and re-captioned with our physics-aware pipeline in Sec. 3.1. In this way, language connects model diagnosis, data indexing, and targeted data acquisition, allowing the training distribution to be adapted to the generator’s observed physical weaknesses.

3.3. PhysThinker for Physics Reasoning

The VLM captioner and upsampler introduced in Sec. 3.1 are initially instantiated with proprietary models. While these models provide strong physics reasoning capability, relying on them for large-scale video re-captioning and inference-time upsampling incurs high monetary cost.

To address this issue, we replace the costly GPT-based captioner and upsampler by distilling their physics reasoning capabilities into two efficient 4B vision-language models: **PhysThinker-C** for physics-aware video captioning and **PhysThinker-U** for inference-time prompt upsampling. PhysThinker-C learns from GPT-generated video-to-text annotations, while PhysThinker-U is trained to expand short text or image-text conditions into physics-rich prompts. Together, they provide scalable and low-cost alternatives to commercial models for large-scale annotation and inference. Training details of our PhysThinker are provided in App. B.5.3.

4. Benchmark and Critic

4.1. Critics for Physics Caption

Fig. 3(b) illustrates the overview of our physics-aware critic used in the agentic loop. To systematically evaluate whether a video caption faithfully and comprehensively represents the physical dynamics in a video, inspired by the caption evaluation protocol used by Cosmos 3 [73], we adopt **precision** and **recall** on the atomic assertions as critics, and specialize both metrics to physical content. Precision evaluates whether generated physical claims are visually supported by the video, while recall measures coverage of salient physical processes.

For evaluation of **precision**, the critic first decomposes the complete generated caption into atomic and independently verifiable claims. The critic is then given the full video together with each atomic claim and classifies it as *correct*, *incorrect*, or *uncertain*. We compute micro-averaged precision as $\text{Precision} = N_{\text{correct}} / (N_{\text{correct}} + N_{\text{incorrect}})$. **Recall** instead focuses specifically on physical dynamics. The critic receives the generated caption together with a set of human-curated atomic physical assertions and determines whether each ground-truth assertion is sufficiently covered by the caption. An assertion receives a positive match only when the caption explicitly states or unambiguously entails its complete physical meaning. Recall is therefore computed as $\text{Recall} = N_{\text{pass}} / N_{\text{ground-truth}}$. We finally report the F1 score, defined as the harmonic mean of precision and recall, $F1 = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall})$, as an overall measure of caption quality.

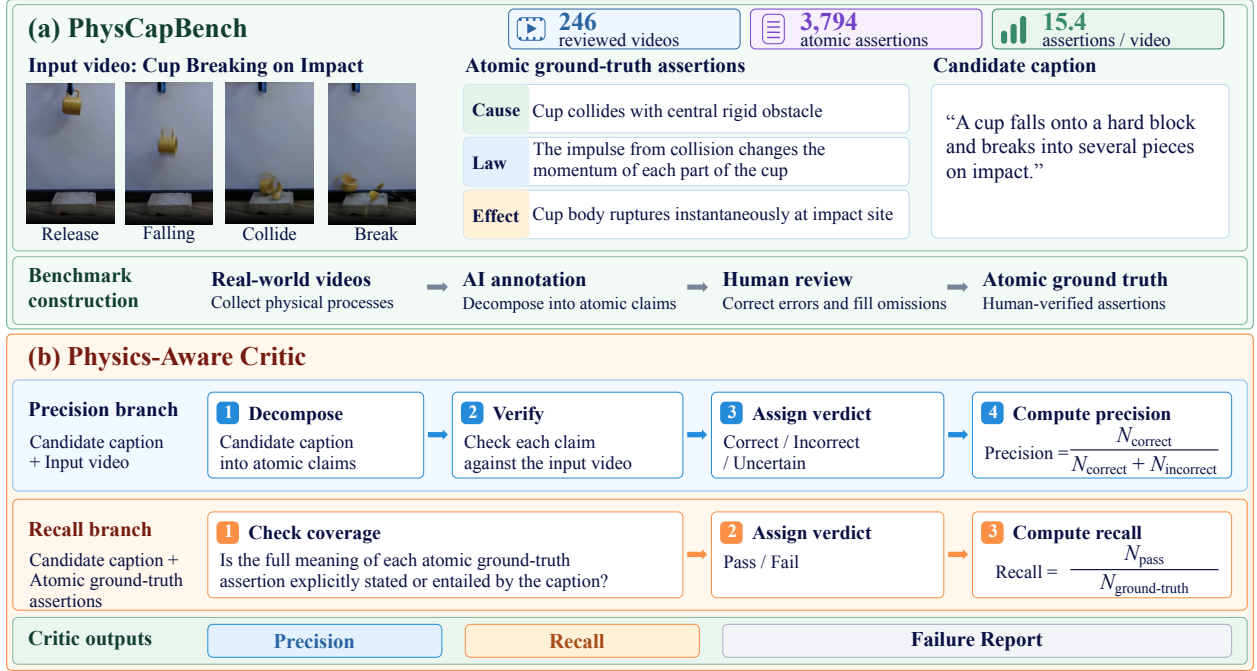


Figure 3 | **Overview of our physics-aware critic and physics caption benchmark.** The critic evaluates generated captions from complementary precision and recall perspectives by verifying atomic claims against the input video and measuring the coverage of human-curated physical assertions.

4.2. Physics Caption Benchmark

Based on the critic, we further introduce **PhysCapBench** (Fig. 3(a)), a dedicated benchmark for evaluating whether video captioning models can faithfully and comprehensively describe salient physical phenomena and their underlying dynamics, consisting of 246 physics-rich videos collected from Physics-IQ [3] and YouTube. We construct fine-grained physical annotations for each video using an agent-assisted, human-verified pipeline. AI annotators identify salient physical processes and decompose them into atomic assertions describing causes, relevant physical laws and effects. All candidate annotations are subsequently reviewed by humans, who remove incorrect or ambiguous assertions and supplement missing ones when necessary.

The final PhysCapBench contains **3,794 human-verified physical assertions** over 246 videos, with an average of 15.4 assertions per video. These annotations provide fine-grained supervision for evaluating whether a caption captures the key physical dynamics and reasoning in each video. Detailed implementations are provided in App. B.1, and benchmark samples are listed in App. E.

5. Experiment

5.1. Experimental Details

Dataset and Backbones. At the core of Physis-Lang is the construction of a physics-enriched video-language training dataset comprising 183K real-world videos. It combines 71K high-quality samples filtered from WISA-80K [74] with 112K additional videos selected through our deficiency-guided retrieval pipeline (Sec. 3.2). We use the dataset to fine-tune several existing video-generation backbones without modifying their architectures and training objectives. Specifically, we consider Wan2.1-14B [75], for which the T2V-14B and I2V-14B variants are fine-tuned separately, and the Cosmos3 family [73], including Edge (4B), Nano (16B), and Super (64B), whose unified backbone supports both T2V and I2V generations. Further implementation details are provided in App. B.

Baselines and Evaluation. Our comparison involves cutting-edge pretrained models, including CogVideoX1.5-5B [76], Wan2.2-TI2V-5B [75], HunyuanVideo-1.5 [77], and the closed-source Veo 3.1. We also compare with

Table 1 | Physical fidelity on Physics-IQ Verified.

Model	ST	S	WS	MSE	Final
CogVideoX1.5-5B [76]	17.41	34.49	21.71	16.18	22.45
Wan2.2-TI2V-5B [75]	23.26	33.89	23.18	23.19	25.88
HunyuanVideo-1.5 [77]	22.80	47.10	30.46	26.05	31.60
Veo 3.1	19.01	52.16	35.45	33.32	34.99
Kandinsky-WM [79]	25.69	36.37	17.22	12.69	22.99
Self-Refinement [25]	22.49	37.93	23.25	25.13	27.20
PhiZero [30]	56.79	37.01	27.59	42.29	<u>40.91</u>
Cosmos3-Nano	29.38	51.76	38.92	40.87	40.23
Ours (Cosmos3-Nano)	33.59	55.38	42.45	42.23	43.41

Table 2 | Physical adherence on PhyGround.

Model	General	Physics	Overall
CogVideoX1.5-5B [76]	59.02	58.36	58.68
Wan2.2-TI2V-5B [75]	62.94	62.46	62.70
HunyuanVideo-1.5 [77]	67.14	65.42	66.28
Veo 3.1	71.92	66.54	<u>69.24</u>
Kandinsky-WM [79]	58.96	57.28	58.12
Self-Refinement [25]	58.54	57.88	58.22
PhiZero [30]	52.27	63.42	57.85
Cosmos3-Nano	65.46	64.90	65.18
Ours (Cosmos3-Nano)	71.54	68.26	69.90

Table 3 | Physical correctness on PhyGenBench.

Model	Force	Light	Heat	Material	Overall
CogVideoX1.5-5B [76]	38.33	57.33	55.56	43.33	48.75
Wan2.2-TI2V-5B [75]	40.83	60.67	43.33	40.83	47.50
HunyuanVideo-1.5 [77]	47.50	56.67	44.44	41.67	48.33
Veo 3.1	65.00	70.67	57.78	65.83	<u>65.63</u>
PhysVid [26]	48.33	62.00	33.33	40.83	47.92
PhyGDPO [78]	43.33	62.67	43.33	41.67	48.96
Self-Refinement [25]	46.67	59.33	44.44	42.50	49.17
Cosmos3-Nano	64.17	64.00	57.78	59.17	61.67
Ours (Cosmos3-Nano)	67.50	72.67	75.56	69.17	71.04

Table 4 | Physical commonsense on VideoPhy-2.

Model	SA	PC	SA $\geq$ 4	PC $\geq$ 4	All	Hard
CogVideoX1.5-5B [76]	3.74	4.10	63.79	74.79	49.41	33.15
Wan2.2-TI2V-5B [75]	3.69	4.39	60.91	85.45	53.98	42.13
HunyuanVideo-1.5 [77]	3.84	4.57	67.85	92.39	62.94	51.69
Veo 3.1	4.39	4.06	84.43	80.54	68.87	<u>58.43</u>
PhysVid [26]	3.50	4.27	55.33	82.74	48.22	29.78
PhyGDPO [78]	3.74	4.64	62.27	93.57	59.56	44.94
Self-Refinement [25]	3.60	4.26	56.01	82.91	47.88	28.09
Cosmos3-Nano	3.86	4.49	66.33	90.19	60.41	48.31
Ours (Cosmos3-Nano)	4.09	4.44	75.97	89.00	<u>68.02</u>	62.36

other methods for promoting physical realism, including PhysVid [26], PhyGDPO [78], Self-Refinement [25], Kandinsky-WM [79], and PhiZero [30]. The evaluation is conducted on four widely used physical benchmarks: VideoPhy-2 [2] and PhyGenBench [1] for Text-to-Video (T2V) generation, and PhyGround [80] and Physics-IQ Verified [81] for Image-to-Video (I2V) generation. To better distinguish differences in generation quality, we replace the original offline VLM evaluators of VideoPhy-2 and PhyGenBench with GPT-5.5, which provides more discriminative assessments in our comparisons (App. G). We evaluate all models using the same scoring protocol within each benchmark to ensure fair comparisons.

5.2. Comparison with State-of-the-Art Models

Tables 1–4 compare Cosmos3-Nano fine-tuned using our Physis-Lang with cutting-edge generative models and other baselines designed to improve physical realism in video generation. For better comparison, we rescale each benchmark to a 100-point scoring scale. We highlight three key advantages of our method: (i) **Consistent and significant improvement over base model.** Building on pretrained Cosmos3, Physis-Lang further improves its performance across all four evaluated physical video benchmarks. (ii) **State-of-the-art among open-source models.** Our method achieves the strongest performance among all open-source models and methods across the four benchmarks. (iii) **Outperforming the best closed-source model.** Our method surpasses Veo 3.1 on Physics-IQ Verified, PhyGround, and PhyGenBench, and remains competitive on VideoPhy-2, trailing by only 0.85 points on the full set (68.02 vs. 68.87) while outperforming it by 3.93 points on the hard split (62.36 vs. 58.43). Qualitative examples in App. C demonstrate the improved generation quality of our method across the evaluated benchmarks, while App. D further shows our strong performance in driving and robotics scenarios.

5.3. Generalization across Backbones and Scales

Tab. 5 evaluates whether our method generalizes across model families and parameter scales. We draw three conclusions: (i) **Consistent gains across model families.** Our method improves Wan2.1-14B and Cosmos3-Nano-16B by 7.05 and 6.22 points on average, respectively, with gains on all four physical benchmarks. (ii) **Effectiveness across model scales.** Within the Cosmos 3 family, our method yields average gains of 3.24, 6.22, and 5.02 points on Edge-4B, Nano-16B, and Super-64B, respectively. These results demonstrate benefits across a broad range of model sizes. (iii) **Preserved general video quality.** Our fine-tuning strategy does not compromise general video generation quality; please refer to App. H for more details.

Table 5 | **Effectiveness on different backbones.** Each cell shows the improvement brought by our method. Mean Δ is the average improvement across the four benchmarks.

Generator	PhyGenBench	Physics-IQ Verified	VideoPhy-2	PhyGround	Mean Δ
<i>Across model families</i>					
Wan2.1-14B	56.67 $\rightarrow$ 65.83	27.87 $\rightarrow$ 35.15	57.02 $\rightarrow$ 65.65	61.52 $\rightarrow$ 64.64	+7.05
Cosmos3-Nano-16B	61.67 $\rightarrow$ 71.04	40.23 $\rightarrow$ 43.41	60.41 $\rightarrow$ 68.02	65.18 $\rightarrow$ 69.90	+6.22
<i>Across Cosmos scales</i>					
Cosmos3-Edge-4B	48.96 $\rightarrow$ 52.50	32.80 $\rightarrow$ 34.69	32.99 $\rightarrow$ 40.61	66.66 $\rightarrow$ 66.56	+3.24
Cosmos3-Nano-16B	61.67 $\rightarrow$ 71.04	40.23 $\rightarrow$ 43.41	60.41 $\rightarrow$ 68.02	65.18 $\rightarrow$ 69.90	+6.22
Cosmos3-Super-64B	66.04 $\rightarrow$ 70.21	45.92 $\rightarrow$ 50.00	60.91 $\rightarrow$ 72.42	68.67 $\rightarrow$ 68.98	+5.02

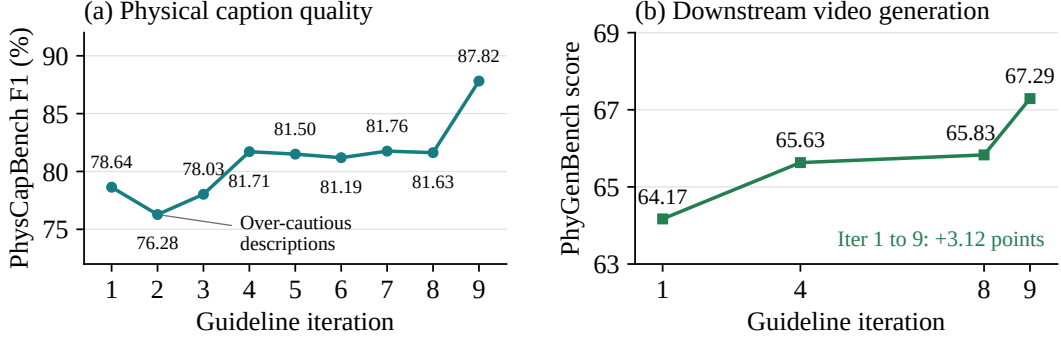


Figure 4 | **From caption evolution to video generation.** Left: PhysCapBench F1 at different iterations. Right: physics-aware video generation performance of the same pretrained Cosmos3 model with inference captions produced at different iterations.

5.4. Ablation and Analysis

5.4.1. Agentic Caption Evolution

Fig. 4 evaluates whether caption evolution improves both physical descriptions and downstream video generation. We highlight two observations: (i) **The agentic loop improves physical caption quality.** On PhysCapBench, the F1 score increases from 78.64 at Iteration 1 to 87.82 at Iteration 9, showing the effectiveness of critic-guided guideline refinement. (ii) **The self-evolving guidelines improve video generation.** Keeping the pretrained Cosmos3-Nano fixed, we change only the inference captions produced using guidelines from Iterations 1, 4, 8 and 9. PhyGenBench scores increase from 64.17 to 65.63, 65.83 and 67.29, respectively. App. A provides the complete evolution trajectory. App. F analyses the effect of *physics_reasoning* derived from our agentic loop.

5.4.2. Language-Guided Data Curation

Language-guided data curation (Sec. 3.2) consistently improves physical generation. Adding retrieved videos to the training set improves benchmarks, for an average gain of 3.01 points (Tab. 6). To further examine whether these gains are consistent across our physical tags, we break down the improvements on VideoPhy-2 by physical category (a sample may belong to multiple categories). The gains are positive across all frequently represented categories, such as chemical processes (+8.00), fracture mechanics (+7.45), and cloth deformation (+7.19 points) (Fig. 5).

5.4.3. Prompting, Fine-Tuning, and Negative Guidance

Tab. 7 highlights three findings: (i) **Physical prompting is effective without fine-tuning.** Our physical reasoning prompt improves the pretrained model from 61.67 to 63.33 (A $\rightarrow$ C), demonstrating a zero-shot gain. Moreover, our agentic loop further improves over the manually designed prompt [78](B $\rightarrow$ C), validating the effectiveness. (ii) **SFT provides additional gains.** With inference prompts fixed, fine-tuning brings further improvement: 3.55 points (C $\rightarrow$ G) and 3.75 points (D $\rightarrow$ H). (iii) **Negative guidance complements SFT.**

Table 6 | **Effect of retrieved training data.** Benchmark scores before and after data expansion.

Benchmark	WISA	+ Retrieved	Gain
PhyGenBench	68.12	71.04	+2.92
Physics-IQ Verified	40.68	43.41	+2.73
VideoPhy-2	64.63	68.02	+3.39
Mean	57.81	60.82	+3.01

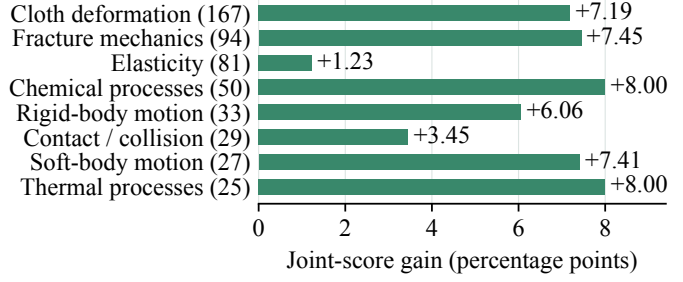

 Figure 5 | **VideoPhy-2 category gains** after retrieval.

 Table 7 | **Prompting and SFT on PhyGenBench.** Base: base caption; M: manually designed physics caption; P: *physics_reasoning*; N: *physics_negative_prompt*.

ID	SFT data	Prompt	Score	ID	SFT data	Prompt	Score
A	None	Base	61.67	E	WISA	Base + P	65.62
B	None	M	61.86	F	WISA	Base + P + N	68.12
C	None	Base + P	63.33	G	WISA + retrieved	Base + P	66.88
D	None	Base + P + N	67.29	H	WISA + retrieved	Base + P + N	71.04

Adding physics negative prompts improves the fine-tuned model by 2.50 (E $\rightarrow$ F) and 4.16 points (G $\rightarrow$ H), respectively.

5.5. Open-Source Deployment with PhysThinker

We train PhysThinker-C and PhysThinker-U for video captioning and inference-time prompt upsampling, respectively. Tab. 8 demonstrates an accuracy–cost trade-off: commercial, captioner-only replacement, and fully local pipelines improve over the pretrained baseline by 7.05, 6.76, and 4.76 points on average, while estimated external annotation costs decrease from \$24.12K to \$0.12K and \$0. Costs cover the re-captioning of training data and the upsampling for benchmarks, whereas accuracy is averaged on four reported benchmarks.

 Table 8 | **PhysThinker replacement on Wan2.1-14B.** VideoPhy-2 reports All / Hard; Cost denotes API expenditure for training-data captioning and benchmark upsampling (USD).

Captioner	Upsampler	PhyGenBench	Physics-IQ Verified	VideoPhy-2	PhyGround	Mean Δ	Cost (USD)
Pretrained Wan2.1-14B		56.67	27.87	57.02 / 43.82	61.52	–	–
Commercial	Commercial	65.83	35.15	65.65 / 59.55	64.64	+7.05	~24.12K
PhysThinker-C	Commercial	65.00	34.71	65.99 / 54.49	64.40	+6.76	~0.12K
PhysThinker-C	PhysThinker-U	63.33	34.13	60.07 / 55.06	64.60	+4.76	0

6. Conclusion

Video world models require physical plausibility beyond visual quality. We introduce *Physis-Lang*, which treats structured physical language as a shared representation across data curation, model training, and inference. PhysCapBench and an agentic self-evolution loop refine physical captions, while language-guided retrieval targets missing physical processes. Experiments with Wan2.1 and Cosmos3 family models across four physical benchmarks of video generation show consistent and significant improvements over the pretrained models. These results show that language, when explicitly structured and self-evolving, can serve not merely as a conditioning interface, but as a practical representation for carrying, refining, and applying physical knowledge in video world models.

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Appendix

A. Prompt Self-Evolution Analysis

As mentioned in Sec. 5.4.1, to better understand how the captioning prompt evolves during iterations, here we visualize the F1-score curve on **PhysCapBench** in Fig. 6. The annotations summarize five observed representative modifications to the prompt design.

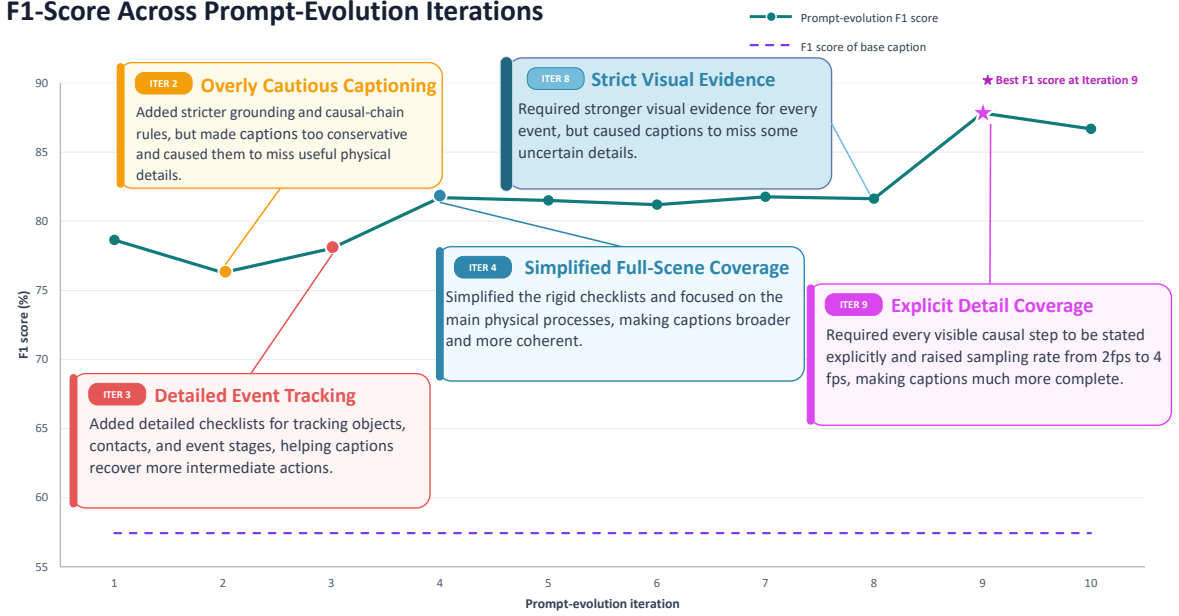


Figure 6 | **F1-score trajectory across prompt self-evolution iterations on PhysCapBench.** The dashed purple line denotes the F1 score of base caption. The five callouts summarize the representative prompt modifications introduced at Iterations 2, 3, 4, 8 and 9. The star identifies the best-performing iteration. The annotations describe the prompt-design changes, whereas all plotted F1 scores are obtained from PhysCapBench.

B. Additional Implementation Details

B.1. Physcapbench and Evaluation Protocol

As illustrated in Sec. 4.2, our PhysCapBench contains 246 videos and includes 3,794 human-verified physical-dynamics assertions with a mean of 15.42 assertions per video. All captions are evaluated by Gemini-3-Flash, using temperature 0.1, a 65,536-token JSON-output limit, and batches of at most 30 claims. Precision decomposes the caption into atomic claims, verifies each against the video, and computes $N_{\text{correct}}/(N_{\text{correct}} + N_{\text{incorrect}})$. Recall denotes the proportion of the 3,794 reference assertions supported by the generated caption.

B.2. More Details on the Agentic Loop

Here we elaborate more implementation details for our agentic loop described in Sec. 3.1. We optimized the physics captions through a generate–critic–revise loop on a fixed development set of 20 videos, which is randomly sampled from our training set, with 273 human-verified atomic physics assertions in total. At each iteration, a fixed GPT-5.5 captioner processed temporally ordered frames and jointly generated *physics_reasoning* and *physics_negative_prompt* in a single call, while only *physics_reasoning* was scored. We use Gemini-3.1-Pro to measure assertion-level recall and video-grounded claim precision, and the evolution agent used both aggregate scores and claim-level failure rationales to revise the prompt.

To construct the upsampling prompt, we preserved the physics-focused requirements of the final caption prompt while adapting its input interface. Instead of describing the visual content from scratch, the upsampler expands the original short benchmark caption, together with visual inputs when required, into the same structured *physics_reasoning* and *physics_negative_prompt* fields.

B.3. Long Context Support on Wan

In Sec. 5.1, one of the backbones going through our fine-tuning is Wan2.1 [75]. The text encoder UMT5-XXL [82] of Wan2.1 imposes a maximum context length of 512 tokens, which can be restrictive for our physics-aware prompts that contain both detailed descriptions and reasoning. To accommodate arbitrarily long captions without modifying the pretrained text encoder, we adopt a sliding-window strategy. Specifically, an over-length prompt is partitioned into overlapping windows, each within the 512-token limit, and the resulting text representations are aggregated to provide conditioning for the video generation model. In practice, the overlap between two consecutive context windows is set to 192 tokens, and we take the average of the embeddings from the two windows over the overlapping region, as shown in Fig. 7.

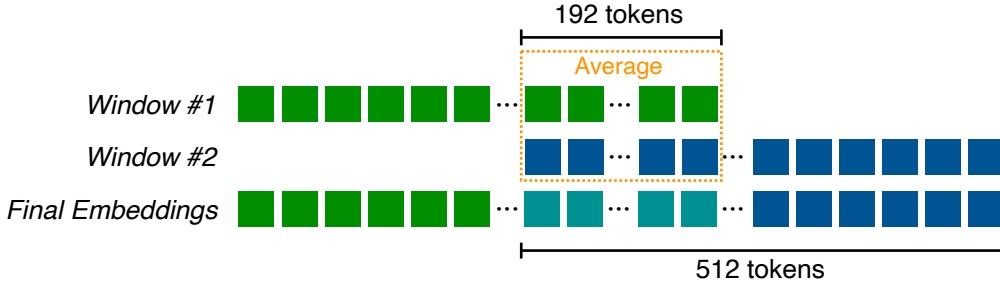


Figure 7 | Sliding-window mechanism for text encoding in Wan2.1.

B.4. More Details on Physics Video Generation Benchmarks

In Sec. 5, we evaluate models on four benchmarks: PhyGenBench and VideoPhy-2 for Text-to-Video generation, and Physics-IQ Verified and PhyGround for Image-to-Video generation. In this section, we briefly describe the evaluation protocol and the physical capabilities assessed by each benchmark.

PhyGenBench. The 160 prompts of PhyGenBench can be divided into four categories of physical phenomena: *force*, *light*, *heat*, and *material*. We calculate the video-level score with the official scripts, but replacing the

evaluator with GPT-5.5. Each video is graded on a 0–3 scale, where 0 indicates a complete fantastical physical behavior and 3 indicates strong physical realism. The final score is obtained by averaging across all 160 samples.

VideoPhy-2. VideoPhy-2 evaluates generated videos from two aspects: *semantic consistency* (SA) and *physical correctness* (PC), each is graded on a 0–5 scale. Its test set contains 591 samples covering common physical phenomena in everyday scenarios, with 178 challenging samples selected to form a Hard subset. We replace its offline VLM evaluator with GPT-5.5 (App. G). The joint accuracy measures the fraction of videos that simultaneously satisfy both $SA \geq 4$ and $PC \geq 4$.

Physics-IQ Verified. Physics-IQ (verified version) evaluates generated videos across multiple physical concepts using four complementary metrics: *spatial consistency* (S), *spatial-temporal consistency* (ST), *weighted-spatial consistency* (WS), and *mean squared error* (MSE). It contains 66 physical scenarios, with three perspectives evaluated per scenario (198 samples in total). It provides ground-truth videos for reference. The final score is computed by aggregating the scores across these metrics and averaging over all evaluated samples.

PhyGround. PhyGround evaluates each generated video using *semantic adherence* (General) and *physical temporal validity* (Physics). Each criterion is scored on a 1–5 scale, and averaged across all 250 samples.

B.5. Training Details

Here we offer training details for Cosmos3, Wan2.1 and our PhysThinker in Sec. 5.

B.5.1. Cosmos 3 family

Cosmos3-Nano. We fine-tuned Cosmos3-Nano using the dataset of about 183K video-caption pairs (Sec. 5.1), including 71K WISA-80K samples retained after filtering for motion and aesthetic quality and removing multi-shot or metadata-incomplete videos, plus 112K videos selected through our deficiency-guided retrieval data pipeline. Each video was paired with a caption combining the base visual description and the generated physics-reasoning annotation. Videos were processed at a spatial resolution of (256×256) , with a maximum caption length of 4,096 tokens and a maximum packed sequence length of 45,056 tokens. During training, the conditioning format was sampled as 70% text-to-video, 20% image-to-video using the first frame, and 10% video-to-video using the first five frames; classifier-free guidance dropout was set to 0.1. We applied LoRA only to the query, key, value, and output projections of the attention layers, using rank 16 and scaling factor 32. Training used bfloat16 precision, fully sharded data parallelism, and full activation checkpointing across 16 GPUs on two nodes. Each GPU processed one packed sequence at a time, with four gradient-accumulation steps, giving a nominal global batch size of 64 packed sequences per optimizer update. We optimized the model with fused AdamW using a learning rate of (5×10^{-4}) , $(\beta_1 = 0.9)$, $(\beta_2 = 0.95)$, $(\epsilon = 10^{-6})$, no weight decay, and gradient clipping at 0.1. The checkpoint at iteration 1,340 was used for the reported benchmark evaluation.

Cosmos3-Super. We fine-tuned Cosmos3-Super on the same training set and caption format described above. Videos were resized and center-cropped using aspect-ratio-dependent buckets at the 256-resolution tier (e.g., 256×256 for square videos and 320×192 for landscape 16:9 videos), with a maximum caption length of 4,096 tokens and a maximum packed sequence length of 45,056 tokens. The conditioning format was sampled as 70% text-to-video, 20% image-to-video using the first frame, and 10% video-to-video using the first two latent frames, corresponding to five video frames under temporal compression by a factor of four. Classifier-free guidance dropout was set to 0.1. We applied LoRA only to the query, key, value, and output projections of the generation expert’s attention layers, using rank 16 and scaling factor 32. Training used bfloat16 precision, fully sharded data parallelism, and full activation checkpointing across 16 GPUs on four nodes, with context-parallel degree 4 and data-parallel degree 4. Each data-parallel replica processed one packed sequence per microstep, distributed across its four context-parallel ranks. With two gradient-accumulation steps, this

yielded a nominal global batch size of eight packed sequences per optimizer update. We used fused AdamW with a learning rate of 5×10^{-4} , $\beta_1 = 0.9$, $\beta_2 = 0.95$, $\epsilon = 10^{-6}$, zero weight decay, and gradient clipping at 0.1. The checkpoint at iteration 1000 was used for evaluation.

Cosmos3-Edge. We fine-tuned Cosmos3-Edge from its pretrained initialization on the same training set and caption format described above. Videos were processed using aspect-ratio-dependent buckets at the 480-resolution tier (e.g., 832×480 for landscape 16:9 videos), with at most 121 frames per video and a maximum caption length of 8,192 tokens. Training used sample-count-based packing with at most two samples per packed batch and a model token capacity of 90,112. The conditioning format was sampled as 70% text-to-video and 30% image-to-video using the first frame; video-to-video conditioning was not used. Classifier-free guidance dropout was set to 0.1. We applied LoRA only to the query, key, value, and output projections of the generation expert’s attention layers, using rank 16 and scaling factor 32. The main training run used bfloat16 precision, fully sharded data parallelism, and full activation checkpointing across 32 GPUs on eight nodes, with context-parallel degree 1, data-parallel degree 32, and no gradient accumulation. This corresponds to a nominal global batch size of 64 samples per optimizer update when every packed batch contains two samples. We used fused AdamW with a learning rate of 5×10^{-4} , $\beta_1 = 0.9$, $\beta_2 = 0.95$, $\epsilon = 10^{-6}$, zero weight decay, and gradient clipping at 0.1. Checkpoints were evaluated at 500-update intervals; the checkpoint at iteration 2,000 was used for the reported benchmark evaluation.

B.5.2. Wan2.1

We fine-tuned both Wan2.1-T2V-14B and Wan2.1-I2V-14B using the identical dataset described in B.5.1. Each caption contains both the base visual description and the generated physics-reasoning annotation. During training, classifier-free guidance dropout was set to 0.1. We applied LoRA only to the query, key, value, and output projections of the generation expert’s attention layers, using rank 256 and scaling factor 512, while the patch embedder and output modules are fully fine-tuned. Training used bfloat16 precision, fully sharded data parallelism, and full activation checkpointing across 16 NVIDIA H100 GPUs. The global batch size is set to 32. We optimized the model with fused AdamW using a learning rate of (5×10^{-4}) with 1000 warmup steps, $(\beta_1 = 0.9)$, $(\beta_2 = 0.95)$, $(\epsilon = 10^{-6})$, no weight decay, and gradient clipping at 0.1. The checkpoint at iteration 8,000 was used for the reported benchmark evaluation.

B.5.3. PhysThinker

PhysThinker consists of two Qwen3-VL-4B-Instruct models fine-tuned with our training data, differing only in the conditioning modality. The captioner is trained on videos paired with GPT-generated physics captions, while the upsampler is trained on short text prompts, with or without a conditioning image, paired with GPT-expanded physics-rich prompts. We applied LoRA with rank 8, scaling factor 32, and dropout 0.05 to all linear projections of the language model, while the entire vision stack (ViT and aligner) is fully fine-tuned. Training used bfloat16 precision, FlashAttention-2, DeepSpeed ZeRO-3 with optimizer and parameter offload to host memory, and full activation checkpointing including the vision tower. Examples are packed to a 57,344-token budget, with video sampled at 4 fps up to 64 frames. The global batch size is 256 packed sequences. We optimized with fused AdamW at a learning rate of (1×10^{-4}) for the language-model adapters and (2×10^{-5}) for the vision tower and aligner, using a cosine schedule with 5% linear warmup, $(\beta_1 = 0.9)$, $(\beta_2 = 0.95)$, $(\epsilon = 10^{-8})$, weight decay 0.1, and gradient clipping at 1.0. Both models were trained for 3 epochs.

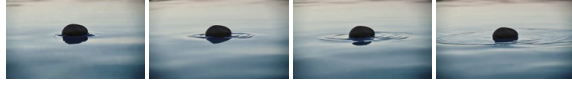
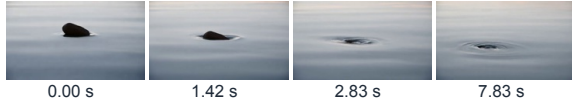
The video-conditioned reasoner takes a video alone and was trained across 128 A100 GPUs. The checkpoint at step 650 was used for the reported benchmark evaluation. The prompt upsampler expands a short generation prompt, and is trained as a two-mode mixture: text-only, and text together with the clip’s first frame. Each corpus contributes to both modes, trained across 32 A100 GPUs. The final checkpoint at step 258 was used for the reported benchmark evaluation.

C. More Qualitative Comparisons

As a supplement to Sec. 5.2, here we show the qualitative examples across the evaluated benchmarks.

PhyGenBench: stone sinking

Official prompt: A stone is gently placed on the surface of a pool filled with water.

Baseline

Ours


physics_reasoning: "The stone **weight remains larger than the buoyant force** because a typical stone is denser than water. **After the support releases the stone, gravity accelerates the unsupported stone downward** into the pool. The stone **penetrates through the water surface** instead of remaining on top of the surface."

physics_negative_prompt: "The stone **remains floating motionless on top of the pool surface after release.**"

Physics-IQ Verified: domino propagation

Official prompt (excerpt): The platform rotates clockwise and the wooden stick hits the first block as it rotates.

Baseline

Ours

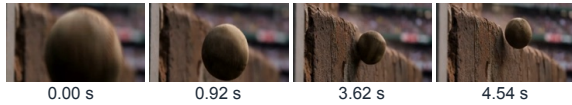

physics_reasoning: "The falling purple block **contacts the right side of the upright blue block**. The purple-blue collision **transfers impulse and angular momentum** from the purple block to the blue block. The blue block **begins rotating leftward about its lower edge** while table friction limits bottom sliding."

physics_negative_prompt: "The colored blocks **topple in separate directions instead of transferring leftward collision impulses along the row.**"

VideoPhy-2: baseball-wall contact

Official prompt: A baseball is thrown with great force; it hits a brick wall and leaves a visible mark.

Baseline

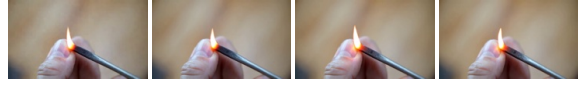
Ours


physics_reasoning: "At impact, the baseball **contacts the brick wall over a small area**. The brick wall exerts a **large normal force on the baseball opposite the baseball's incoming velocity**. The baseball exerts an equal and opposite force on the brick surface through **Newton's third law**. The short collision time produces a **large impulse that sharply reduces the baseball's forward momentum**."

physics_negative_prompt: "The baseball **passes through the brick wall without slowing or transferring momentum.**"

PhyGenBench: potassium flame

Official prompt: A piece of potassium is ignited, emitting a vivid and unique flame as it burns steadily.

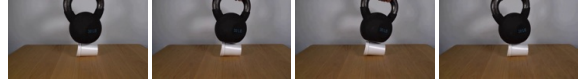
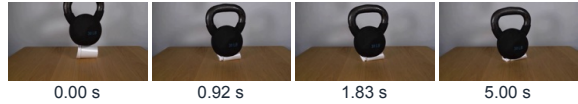
Baseline

Ours


physics_reasoning: "Ignition **supplies heat** to the exposed potassium surface. ... A vivid **violet-lilac flame forms** at the potassium surface and remains attached to the burning region. The flame color comes from **excited potassium atoms and ions emitting characteristic visible wavelengths as the excited electrons lose energy**."

physics_negative_prompt: "The flame burns as a **steady orange candle-shaped plume with no potassium-colored violet-lilac emission**."

Physics-IQ Verified: load-induced cup collapse

Official prompt: A 30lb kettlebell is slowly lowered onto a white styrofoam cup placed on a wooden table on its side. Static shot with no camera movement.

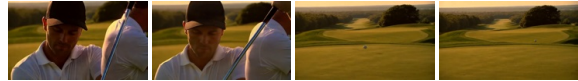
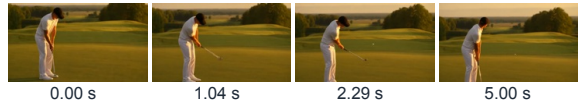
Baseline

Ours


physics_reasoning: "The kettlebell **applies a large downward contact force** to a small region of the styrofoam cup wall. ... The cup material **deforms because the compressive stress exceeds the stiffness and strength** of the thin styrofoam wall. ... The kettlebell center of mass **moves downward as the collapsing cup loses height**."

physics_negative_prompt: "The white cup **collapses downward** while the black kettlebell **remains suspended in place without following the lost support**."

VideoPhy-2: golf chip

Official prompt: A golfer chips the ball onto the green, the ball rolling smoothly towards the hole.

Baseline

Ours


physics_reasoning: "The clubhead **contacts the ball during the swing**. The clubhead applies a **short, strong impulse** to the ball at impact. **Newton's third law** makes the ball apply an equal and opposite impulse back on the clubhead. The impulse **changes the ball from rest to forward and upward motion**."

physics_negative_prompt: "The golf ball **launches into the air before the clubhead contacts it**. The ball flies in a **straight horizontal line without downward curvature from gravity**."

Figure 8 | **Qualitative comparisons on different physics benchmarks.** Each case compares frames generated by the Cosmos3-Nano base model using base caption (Baseline) with those generated by our trained model using base caption augmented with our *physics_reasoning* and *physics_negative_prompt* (Ours) at identical timestamps. Official prompts and selected physics reasoning and negative-prompt excerpts are shown, with bold phrases highlighting physical constraints. Negative excerpts describe violations to avoid. All frames are uncropped.

D. Demos in Driving and Robotics Scenarios

As stated in Sec. 5.2, here we offer 6 demos to illustrate the strong performance of our method in driving and robotics scenarios.

Driving

Approaching queued traffic

Caption excerpt: "The police car travels forward along a suburban paved road toward a signalized intersection."



physics_reasoning: "As the police car **nearns the queue**, the police car decelerates to avoid the stopped vehicles. **Brake force** at the police car tires opposes the car's forward motion. The braking force **reduces the police car forward momentum over time** rather than stopping the car instantaneously."

physics_negative_prompt: "The white van **snaps from forward travel to a complete stop with no braking distance** while surrounding traffic continues through it."

Congested city traffic

Caption excerpt: "A large bus farther ahead remains aligned with the same traffic queue."



physics_reasoning: "Each moving vehicle is constrained by **tire-road friction and steering geometry**, so each vehicle **follows the road direction rather than sliding sideways**. Engine force and braking force **balance closely in congestion**, so the vehicles have small accelerations and very low speeds."

physics_negative_prompt: "The gray sedan and bus **slide sideways across the lane without tire rotation or steering**."

Driving through snowfall

Caption excerpt: "The vehicle travels forward along a snow-covered urban road."



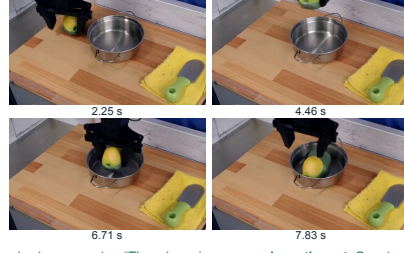
physics_reasoning: "Heavy snow falls through the air across the whole view. **Gravity accelerates unsupported snowflakes downward**. Air drag limits each snowflake's speed and makes the flakes **drift and flutter instead of dropping as rigid projectiles**."

physics_negative_prompt: "The falling snow **moves upward from the road into the sky** while the vehicle travels forward."

Robotics

Placing an avocado in a pot

Caption excerpt: "The robotic arm positions the toy avocado over the pot opening."

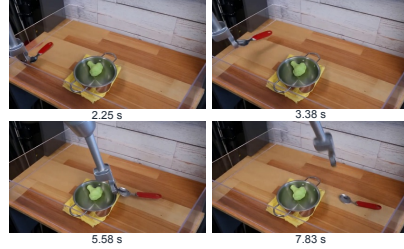


physics_reasoning: "The gripper jaws **open above the pot**. Opening the jaws removes the inward normal forces on the toy avocado. Removing the jaw contact **removes the frictional support** that held the toy avocado against gravity. The toy avocado becomes unsupported over the pot interior. **Gravity accelerates the toy avocado downward** into the stainless steel pot."

physics_negative_prompt: "The green toy avocado **hangs motionless in the air after the claw opens above the pot**."

Transferring a spoon

Caption excerpt: "The spoon moves laterally from the left side of the pot toward the right side of the pot."

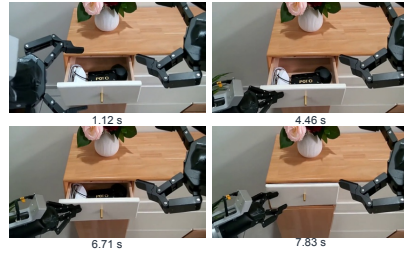


physics_reasoning: "The countertop **contacts the underside of the spoon** when the spoon reaches the surface. ... **Static friction between the spoon and the wooden countertop resists residual horizontal motion**. The spoon **decelerates to rest to the right of the pot**."

physics_negative_prompt: "The released spoon **remains hovering** to the right of the pot instead of settling onto the wooden countertop."

Closing a drawer

Caption excerpt: "The drawer rails constrain the drawer to nearly horizontal translation."



physics_reasoning: "During the closing phase, the left robotic hand **applies a sustained horizontal inward contact force** to the drawer front or front edge. The drawer applies an **equal and opposite contact reaction force** back on the left robotic hand according to **Newton's third law**. The inward contact force **accelerates the drawer along its slide direction toward the cabinet**."

physics_negative_prompt: "The white drawer **slides closed while the left robotic hand never contacts the drawer front or edge**."

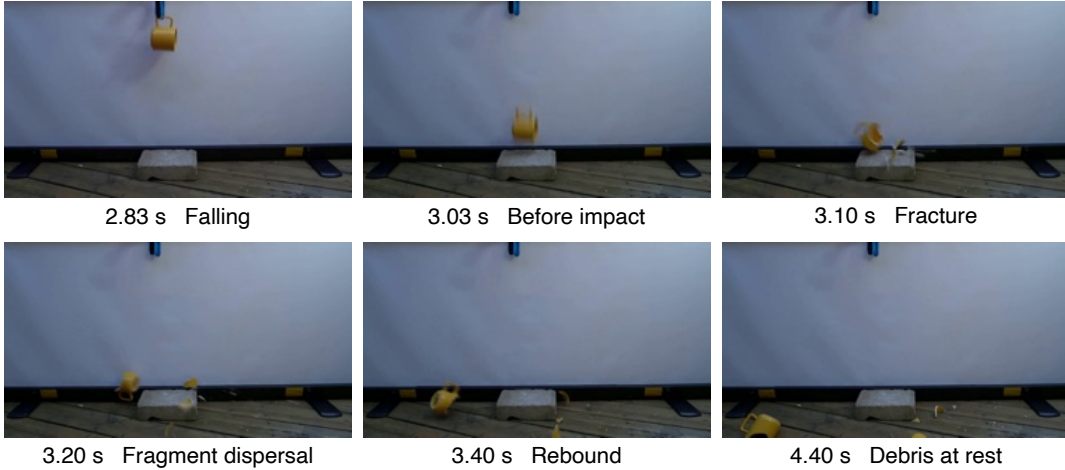
Figure 9 | **Qualitative results on driving and robotics.** All sequences are generated by our trained model using base caption augmented with our *physics_reasoning* and *physics_negative_prompt*. Each example includes four uncropped video frames with timestamps, a caption excerpt, and selected physics reasoning and negative-prompt excerpts.

E. More Examples of PhysCapBench

We provide two examples from PhysCapBench (Sec. 4.2): a falling cup that fractures upon impact and a Newton’s cradle exhibiting coupled collisions. Each example includes six chronological video frames and its complete set of 20 human-curated annotations, categorized as *Cause*, *Law*, or *Effect*. Together, these examples illustrate how the benchmark captures initiating interactions, governing physical principles, and observable outcomes beyond coarse event descriptions.

Cup drop and brittle fracture

8.00 s | 20 assertions

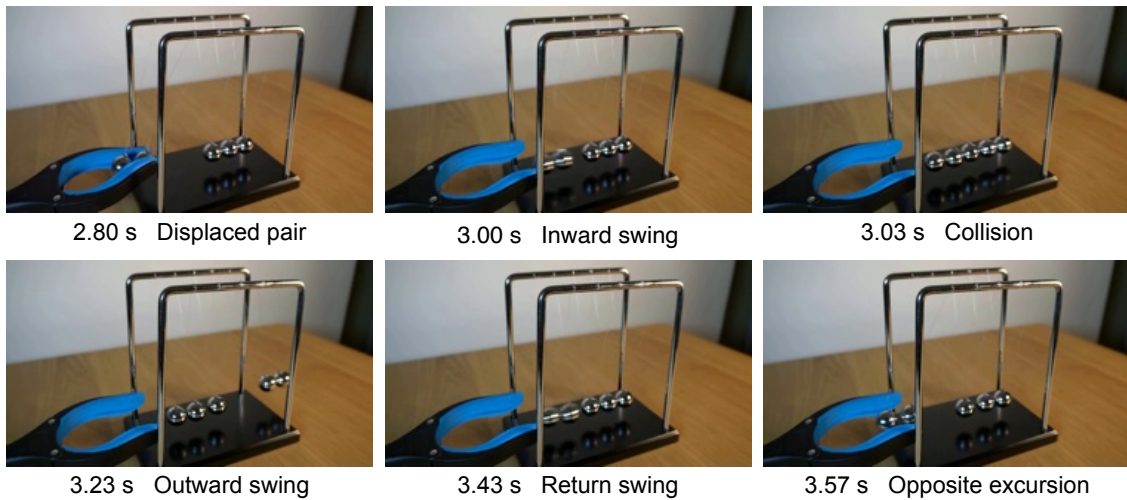


ID	Type	Annotation
01	Effect	Cup body ruptures instantaneously at impact site
02	Cause	The impact force from the obstacle triggers cup body rupture
03	Effect	Cup body breaks into fragments of different sizes
04	Effect	Cup body fragments scatter in multiple directions from impact point
05	Cause	Cup collides with central rigid obstacle
06	Law	The impulse from collision changes the momentum of each part of the cup
07	Effect	The main cup body rebounds to the left side of the frame from the impact point
08	Effect	Small fragments fly away to the right of the obstacle from the impact point
09	Law	The obstacle applies a reverse contact force to the cup body
10	Law	Impact energy converts to material fracture energy
11	Cause	Gravity drives the cup downward
12	Effect	Cup falls downward from top of frame toward central obstacle
13	Law	Cup's falling speed increases with time
14	Law	Cup's gravitational potential energy converts to falling kinetic energy
15	Effect	Cup body separates from obstacle after impact
16	Law	Aerial fragments fall along a parabolic path under gravity
17	Effect	After landing the main cup body slides to the left side of the frame
18	Law	The ground applies a supporting force to the debris
19	Law	Ground friction reduces the fragments' sliding speed
20	Effect	Cup body fragments eventually scatter on both sides of obstacle

Figure 10 | **Cup drop, impact, and fracture.** Six chronological frames accompany the reference physical assertions in their original wording and rank order. Phase labels summarize the displayed frames; assertion IDs are not temporal alignments.

Newton's cradle: coupled collisions

8.00 s | 20 assertions



ID	Type	Annotation
01	Cause	The outer steel ball swinging inward collides with an adjacent steel ball
02	Law	The impulse from collision propagates along the steel ball chain
03	Effect	The opposite outer sphere swings outward after collision
04	Effect	Incident sphere significantly decelerates after collision
05	Law	Contact steel ball experiences an equal and opposite force
06	Law	The middle steel ball array transmits most of the incident momentum
07	Effect	The middle steel ball maintains a small overall displacement
08	Law	Tension in suspension line constrains outer ball to follow arc motion
09	Cause	The outer ball displaced from equilibrium possesses gravitational potential energy
10	Cause	Gravity tangential component drives the outer sphere to swing back
11	Effect	The outer ball swinging downward accelerates toward the lowest point
12	Effect	The outer ball reaches a high speed near the lowest point
13	Law	The outer ball swinging upward decelerates with increasing height
14	Effect	The outer ball changes direction at the lateral high point
15	Effect	The outer balls at both ends periodically alternate approaching the steel ball array
16	Law	Dissipation causes the swing amplitude to gradually decrease
17	Effect	The two balls on the far left swing together to the right
18	Effect	The two balls on the far right swing together to the right
19	Effect	The number of balls lifted on each side is identical
20	Law	Momentum is conserved in all ball collisions

Figure 11 | **Newton's cradle and momentum transfer.** Six chronological frames accompany the reference physical assertions in their original wording and rank order. Phase labels summarize the displayed frames; assertion IDs are not temporal alignments.

F. Analysis on Video Generation Attention Maps

As in Sec. 5.4.1, to measure the effect of *physics_reasoning* derived by self-evolution iterations in Physis-Lang, we use Cosmos3-Nano as the backbone and compare two configurations: inference with base caption only using the pretrained model, and inference with base caption + *physics_reasoning* using the model after SFT on the data we re-captioned with *physics_reasoning* field.

Cosmos3 processes text tokens and video latent tokens in a unified attention sequence. To measure the importance of each token, we compute the attention from sampled visual queries to all text tokens. Specifically, we capture the normalized query and key features after mRoPE and compute the text-conditional attention

$$A(w, q) = \frac{1}{H} \sum_{h=1}^H \text{Softmax}_{w'} \left(\frac{Q_h(q) K_h(w')^\top}{\sqrt{d_h}} \right)_w, \quad (1)$$

where w is the given text token, and q is the video-latent query. H denotes the number of attention heads. The score of a token is obtained by averaging $A(w, q)$ over the uniformly sampled visual tokens, denoising steps, and all transformer layers. For better visualization, we divide each word score by the mean score of all word occurrences in the same prompt. Therefore, the larger the scaled score of a text token is, the greater the role that corresponding word plays in video generation.

Similarly, to obtain the focal regions of an RGB frame with respect to a key sentence S , we average the attention assigned to all text tokens within the span of S :

$$M_S(q) = \frac{1}{|\mathcal{T}(S)|} \sum_{w \in \mathcal{T}(S)} A(w, q), \quad (2)$$

where q is a visual latent token encoded from the RGB frame, and $\mathcal{T}(S)$ denotes the corresponding token set. Each sampled visual query retains its latent coordinate (t, y, x) , allowing the values $M_S(q)$ to be arranged into a spatial map.

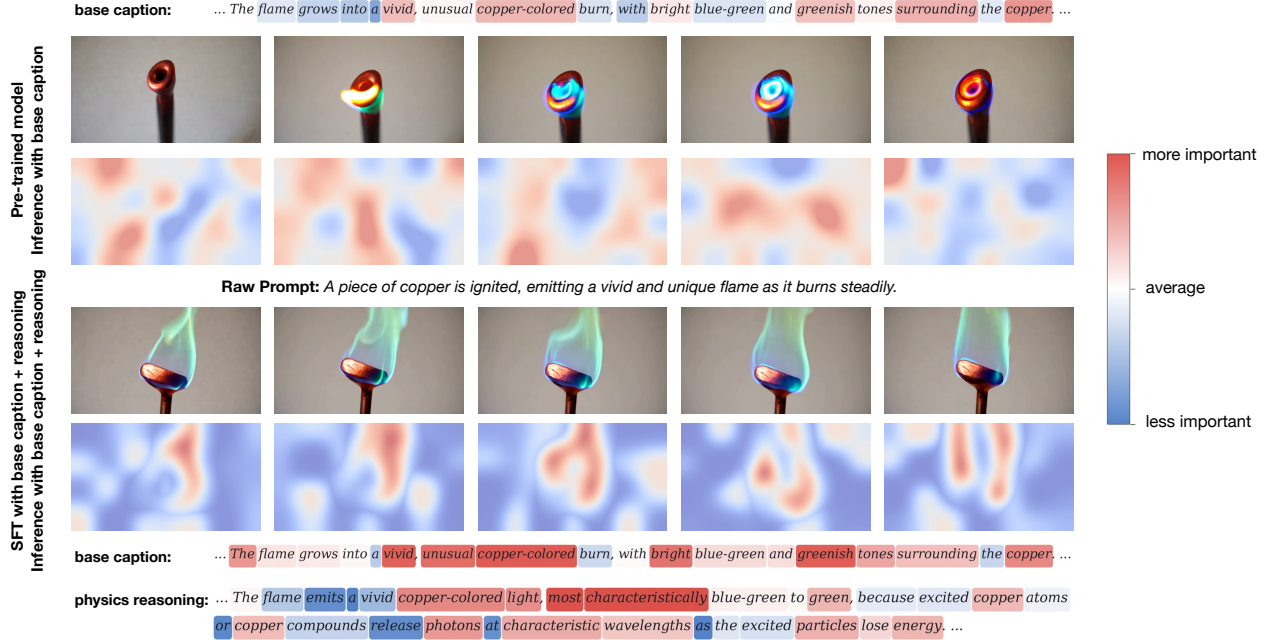


Figure 12 | **Visualization for the attention scores of visual and text tokens.** When adding *physics_reasoning* into the SFT and inference process, the generative model can better capture the keywords for the core physical process, and generate videos with physical realism.

As shown in Fig. 12, when incorporating *physics_reasoning* into the SFT and inference process, the model can better locate the keywords which determine the physical process, like “copper-colored” and “greenish” in

the base caption, and “photons”, “wavelengths” and “energy” in the physics reasoning. For the RGB frames, the region depicting the flame color reaction is also more prominent on the attention map.

Moreover, to examine the important role of our fine-tuning in enabling the model to effectively utilize *physics_reasoning*, we separately feed base caption + *physics_reasoning* into the pretrained model and the fine-tuned model. Tab. 9 lists the relative increase in the attention scores of keywords in *physics_reasoning* after training.

Table 9 | **Relative increase of attention scores of physical keywords after fine-tuning.**

Caption	Δ (%)
...Regions directly under fingertips COMPRESS first and most strongly. ...	19.2
... HEAT TRANSFER lowers the average molecular kinetic energy of the juice, ...	52.2
... GRAVITY still acts downward on all gas and particle mass, ...	22.5
...The change in light speed at the lens surfaces BENDS the rays. ...	48.7

G. Validation of GPT-5.5 as a Better Evaluator

In Sec. 5.1, we replace the offline VLM evaluators for PhyGenBench and VideoPhy-2 with GPT-5.5. In this section, we illustrate the rationale for this replacement using VideoPhy-2 as an example.

As shown in Fig. 13 and Tab. 10, distinct video generative models often receive nearly identical scores from the pretrained VLM, leading to results that do not faithfully reflect their physical performance. In contrast, GPT-5.5 produces assessments that align with human judgment in most cases.

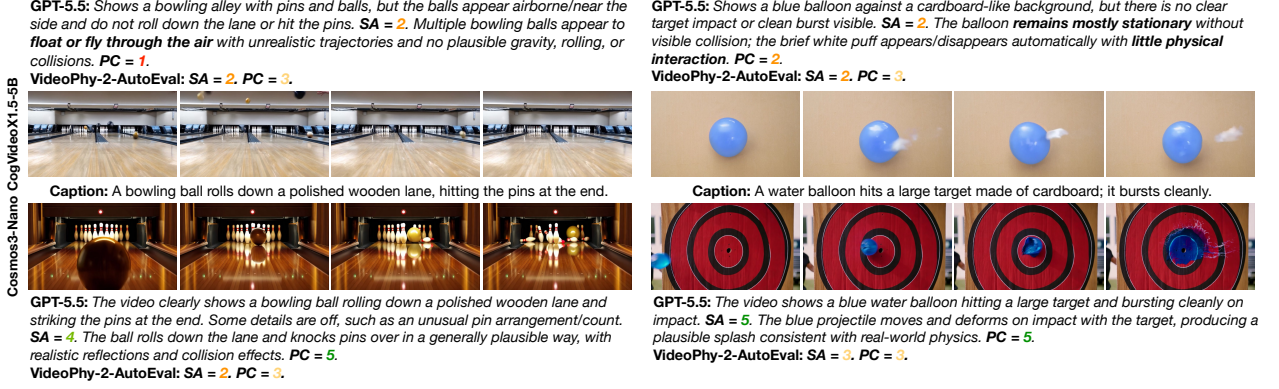


Figure 13 | **Different evaluators for videos generated by CogVideoX1.5-5B and Cosmos3-Nano.** While the auto-evaluator of VideoPhy-2 tends to assign similar mid-range scores to videos with substantially different levels of physical realism, GPT-5.5 provides more discriminative assessments, better reflecting the differences in physical realism across videos.

Table 10 | **Comparison of VideoPhy-2 scores under different evaluators.**

Generative Model	Pretrained VLM	GPT-5.5
CogVideoX1.5-5B	23.01	49.41
Wan2.1-T2V-14B	23.52	57.02
Cosmos3-Nano	22.50	60.41
Veo-3.1	23.69	68.87

H. General Generative Capability of Physis-Lang

In Sec. 5.3, we demonstrate that Physis-Lang explicitly optimizes captions and improves the physical realism of generated videos. A natural question is whether such physics-aware captions could compromise the model’s general capability of video generation. To investigate further, we evaluate the pretrained and Physis-Lang-enhanced models on VBench for the I2V task. As shown in Tab. 11, Physis-Lang maintains comparable performance to the pretrained models, suggesting that the richer and more structured captions introduced by Physis-Lang can provide useful information beyond explicitly physical scenarios.

Table 11 | **Comparison of different models on VBench.**

Model	VBench-I2V
Wan2.1-14B (Pretrained)	86.86
Wan2.1-14B (Physis-Lang)	87.51
Cosmos3-Nano (Pretrained)	88.32
Cosmos3-Nano (Physis-Lang)	88.69